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Innovative Perforation Strategy for Remedial Sand Control in Cemented Monobore Wells

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Abstract

Sand production is recognized as a productivity reducing factor during production phase, especially for marginal reservoirs. Intrusion of sand production at surface requires effort to choke back production below Maximum Sand Free Rate (MSFR) and resulting in sub-optimal production. Uncontrolled sand production at surface could lead to equipment erosion, flow restrictions, and increased of operational risks such as Loss of Primary Containment (LOPC). Effective remedial sand management is crucial for maintaining downhole and surface well performance and its operational efficiency to maximize the gas reserve.

Working in slim hole cemented monobore gas well is truly challenge in many reasons thus requires a unique approach on the conveyance and practical remedial sand control solution to minimize sand production and prolong well life to maximize reserve. A bottom-up completion strategy is adopted as the Reservoir Management Plan (RMP) where many of the remaining reserve and re-completion opportunities are at the shallower sand intervals (TVD 1500m and less), which known as sand prone producing reservoir. These situations will require practical and cost-effective remedial sand management plan for all wells including highly deviated wells. Thorough feasibility study between active technique (mechanical & chemical) and passive remedial options was evaluated to determine the best sand remedial approach to produce well at its optimum gas production rate and delaying the sand entry into the wellbore. Options for chemical and mechanical remedial may seem efficient but introduce more complexity and higher operating cost based on the feasibility study conducted. Implementation of oriented perforation based on geomechanics study recommendation is the practical method as it increases the drawdown operating envelope to produce gas without sand issue as the well is to be perforated parallel (oriented) to the maximum overburden stress.

The selected passive oriented perforation method that carried out in the field demonstrated the best and practical remedial sand control solution in cemented monobore gas well by minimizing and delay sand production while enhancing well productivity and reducing intervention cost. The findings contribute to improved sand management practices, ensuring sustained well performance and operational efficiency in the challenging reservoir environments. This paper will explain on the company journey on evaluating the practical remedial sand control method in Sarawak (SK) gas well, details process involves during the

SK Completion Design - Cemented Monobore Completion

Characterized by a single tubing diameter from the surface to the reservoir, a cemented monobore well presents unique challenges and opportunities in well intervention and stimulation operation. The design eliminates the need for multiple casing strings and liner and simplifies the completion accessories requirement as per example in [Figure 1](#). With this, a cemented monobore completions offer a low-cost solution for marginal gas fields, particularly in high-cost drilling environments and short-life wells. This type of completion has gained traction due to their simplicity (simple completion design), reduced operational complexity (no requirement for wellbore cleanout and packer setting execution) and cost efficiency (capability to perform offline cementing operation which led to shorter rig time).

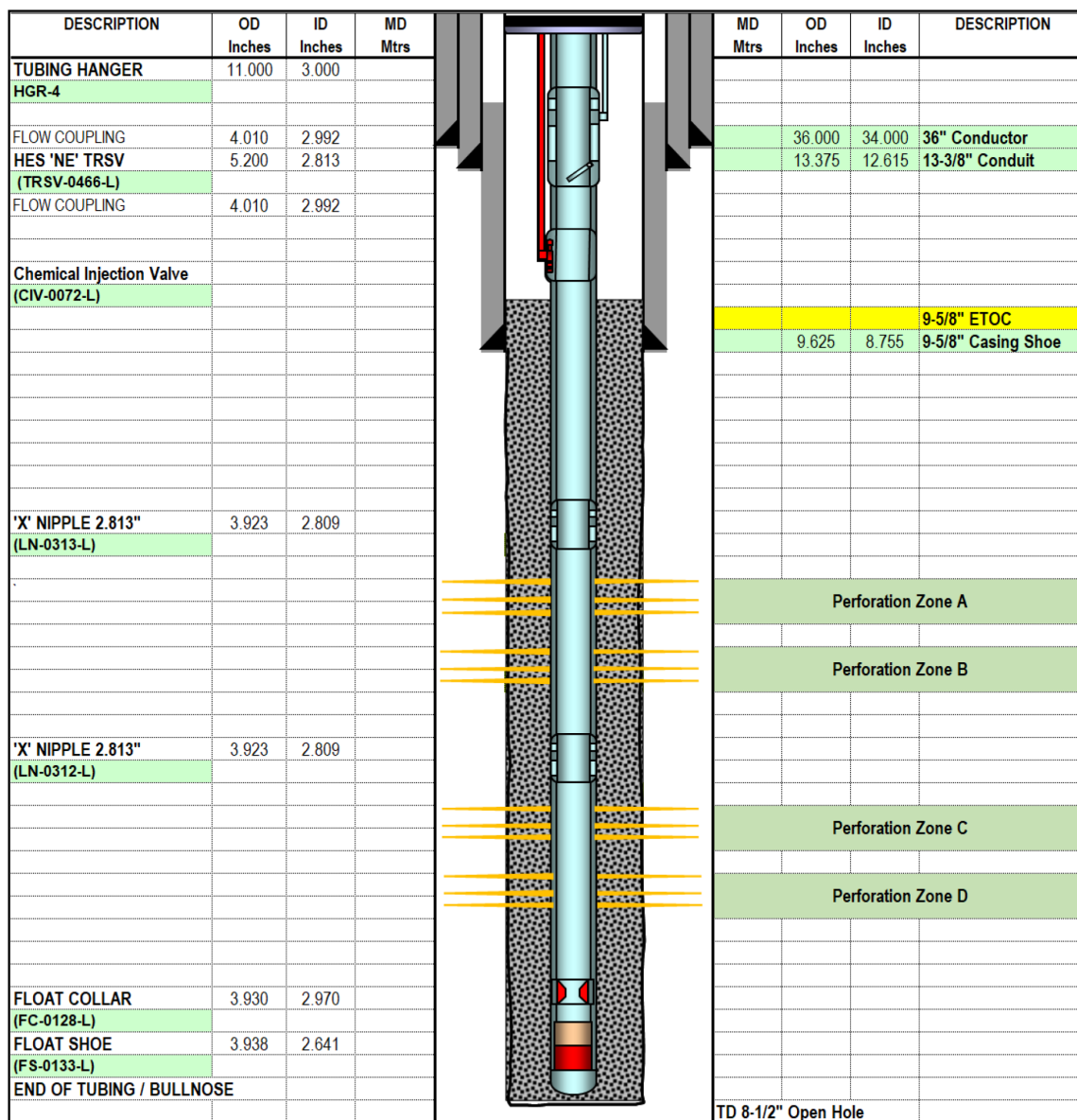


Figure 1—Standard SK Gas Well Cemented Monobore Schematic

On the other hand, the cemented monobore completion also has its downside such as the poor cementing job can lead to complex well accessibility, no sand control, complexity in remedial sand control (uniform diameter limits space for downhole tools deployment) and potential complexity for future Plug & Abandonment (P&A) operation; if have unretrievable fish to access the Caps Rock layer. On the flow assurance view, the management of sand production remains a critical challenge in such configurations, particularly for high-deviation wells with slim-hole diameters. Uncontrolled sand production can damage equipment, obstruct flow pathways, and necessitate frequent interventions, which are impractical on unmanned platforms.

There are 2 types of completions design in SK Field. For Oil Producer (OP), it was completed with Open Hole Expandable Sand Screen (OHES) as the primary sand control measure, allowing hydrocarbons to flow while filtering out sand. Gas wells on the other hand, are completed using 3-1/2" cemented monobore with drawdown control for sand management. There are 2 types of configurations which is 3-1/2" inside 9-5/8" casing & 8-1/2" OH and 3-1/2" tubing inside 7" casing & 6-1/8" OH. The reservoir was completed in a "bottoms-up" strategy to maximize the gas recovery.

- Isolate existing bottom zone and perforate shallower zone
- Commingled existing zone with new perforation of shallower zone

This paper explores the journey of SK Asset Team in implementing a practical remedial sand control in cemented monobore wells in offshore Sarawak, from the understanding of sand issue, evaluating on remedial sand control until the field applications to sustained well productivity.

Sand Management Journey in SK Gas Field

Sand production is a persistent issue in the oil and gas industry, particularly in reservoirs with weakly consolidated formations. In such reservoirs, sand grains become dislodged during production, leading to several operational concerns, including erosion of downhole and surface equipment, reduced production efficiency, and potential well integrity issues. Managing sand production effectively is crucial to maintaining the economic viability and safety of oil and gas operations.

The SK Gas Field presents unique challenges due to varying reservoir conditions and well completion designs. As described in the above cemented monobore section, the real issue since the early production phase is these wells does not complete with any active sand control and only relying on a drawdown control to eliminate sand intrusion into wellbore and up to surface. Over time, some of these gas wells have started producing sand, leading to suboptimal production and increased LOPC risk. In year 2020, SK field was experienced with a severe LOPC incident which served as a learning curve in future sand management in SK Asset as shown in [Figure 2](#).

- Safety: Loss of primary containment (LOPC) due to sand cut.
- Facilities equipment damage: pin hole leak, valve and condensate pumps (mech seals) damage, cut in pump impellers, hydro cyclone liners, plugging equipment and filling vessels, amounting to \$MM in repairs.
- Production downtime: platform require shutdown for months for LOPC investigation, cleaning and repair/replacement.
- Reserves loss: some gas wells are limited to <1 MMscfd and some shut in to prevent sand production to surface.

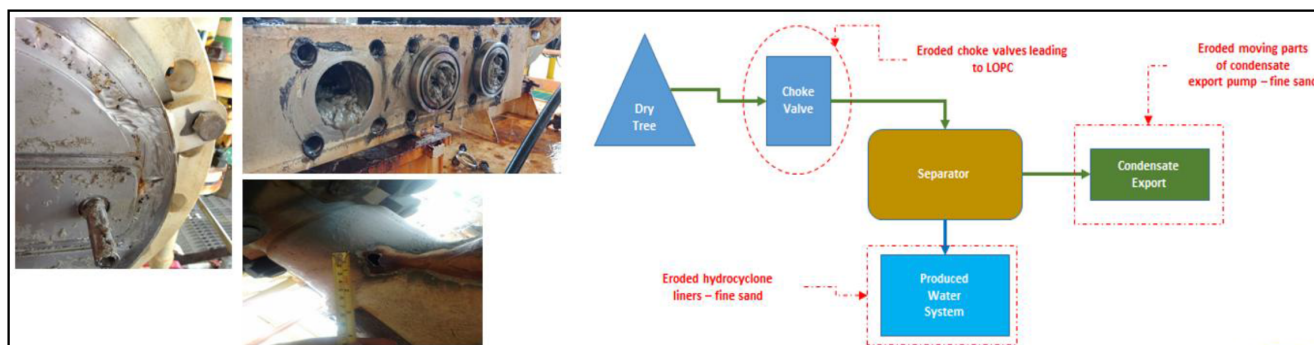


Figure 2—Example of LOPC issue experienced

Sand Management Strategy before 2020

- Monitor/minimize: Sand production has been ‘controlled’ by MSFR (Maximum Sand Free Rate) during well testing. Sand monitored using ultrasonic sand detectors at test header. Despite this, sand has been produced in small quantities and fines carry through to system and result in damage (sand detectors unable to see sand <20 microns).
- Remove/Repair: Sand is removed from the system when required, with repairs as necessary. Costly with potential high downtime.

Sand Management Strategy after 2020

The SK Asset had come out with integrated 3-prong sand management strategy which focused on Downhole (Prevention), Flowline (Protection) and Topsides (Protection).

The strategy is described as follows:

- Prevent: Evaluated downhole remedial sand control and Oriented Perforation strategy was adopted as the practical remedial strategy.
- Monitor: MSFR test monitored via ultrasonic sand detectors (better calibration, link back to KL office for monitoring, and installation of sand detectors on each identified well flowline, sampling, inspection for erosion/metal loss. training for field personnel for better understanding of sand detectors.
- Protect: Equipment installation to protect surface facilities
 - Sand Service chokes on selected wells.
 - Sand jetting system in Production Separator
 - Pump suction strainer periodic cleaning

Based on the MSFR testing conducted on all SK Gas well from Q2 to Q4 of 2020, it was affirmed the correlation of sand production on perforation intervals ranging from 1000m TVD to 1500m TVD.

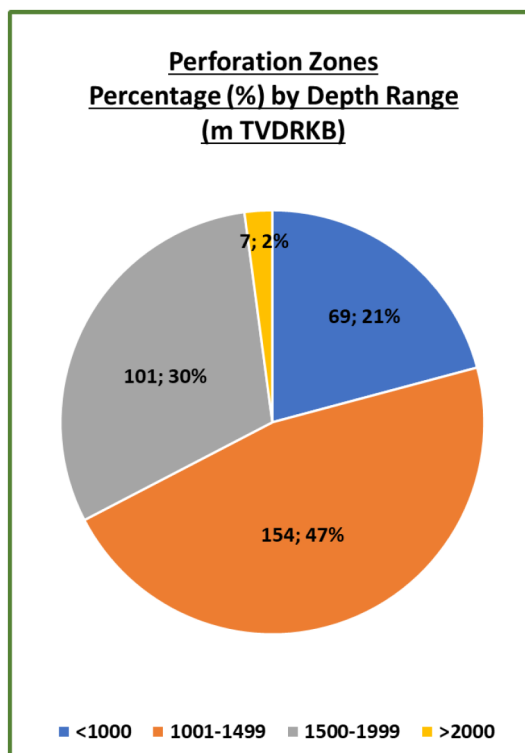


Chart 1—% of the perforated depth

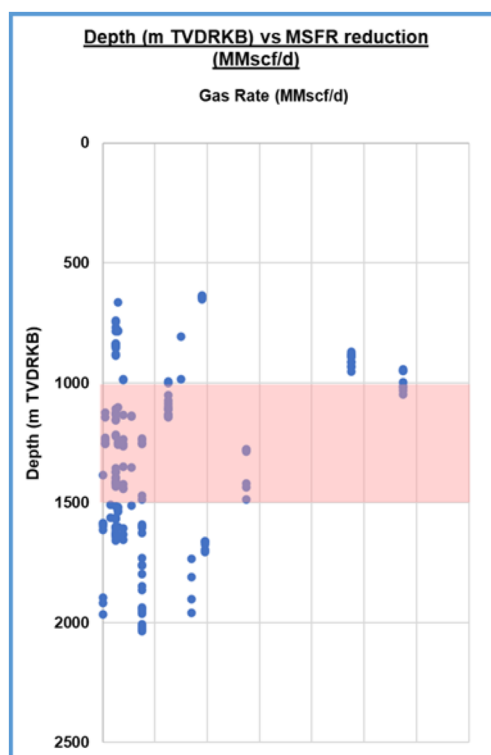


Chart 2—Perforated interval is from 1000 – 1500mTVD

From Well Intervention perspective, the remedial cleanout operations – whether via sand bailing using Slickline (SL), wellbore cleanout using E-Line (EL) with cleaning chamber or sand cleanout via Coiled Tubing (CT) have its own challenges, particularly due to platform constraints and wellbore condition (total sand accumulation inside wellbore).

- SK Gas Platform is an unmanned platform with limited deck space and crane limitation, making it difficult to mobilize a full CT package. CT operations require vessel support to accommodate pumping equipment and provide crew accommodation, adding to logistical complexity and cost.
- The mode of operation at SK Gas Field is daily tripping via Fast Crew Boat (FCB). This restricts the duration of SL sand bailing runs or wellbore cleanout run via EL conveyance, limiting the ability to collect samples and clear obstructions within the available working hours.
- The wellbore cleanout runs via EL conveyance is slow process due to requirement to collect debris / sand until the chamber is full, especially in highly deviated wells where is require a tractor conveyance to the target depth.

With the remaining reserves and future recompletion targets located in shallower intervals, especially in highly deviated wells, there is a growing need for a practical and cost-effective sand management strategy. A thorough feasibility study is essential to evaluate the most suitable remedial approach, balancing technical effectiveness and economic value, while minimizing the risk of sand entry into the wellbore.

Remedial Sand Control Option for Cemented Monobore Wells

A simultaneous work is started between Well Intervention Team and Production Engineering group. Well Intervention focuses on Remedial Sand control option while Production Engineering Team focuses on MSFR and Geomechanics analysis. Evaluating a different remedial sand control technique is critical phase in selecting the suitable approach of sand management journey for SK Gas wells.

The industrial options on Remedial Sand Control in Cemented Monobore is described in [Table 1](#).

Table 1—Remedial Sand Control option in Cemented Monobore Completion

Well Type	Options	Methods	Quick Info
Cemented Monobore	Mechanical	• Through Tubing Gravel Pack • Through Tubing Sand Screen	• Provide sand retention inside wellbore • TTGP have proppant support
	Chemical	• Sand Consolidation • Sand Agglomeration	• Method of stopping sand production by artificially bonding the formation sand grains into a consolidated mass and increase the strength of formation. • Require vessel support operation
	Passive	• Oriented Perforation • Drawdown control	• Shooting in specific direction in the wellbore to minimize sand production • Production rate control (MSFR / MASR) & Bean Up / Down policy

The SK Gas Platform is unmanned and has a smaller deck space. Working in slim hole cemented monobore requires a unique approach to the conveyance and practical remedial sand control to eliminate / minimize sand production and prolong well life. Based on the evaluation conducted, a conventional remedial sand control methods like gravel packing or chemical treatment are often unsuitable in these marginal fields and small platform space due to their complexity, environmental concerns, and high costs. This has created a demand for innovative and cost-effective solutions tailored to slim-hole environments.

The detailed comparison between 4 main methods is described in [Table 2](#).

Table 2—Comparative evaluation of remedial sand control methods

Factor	Methodology			
	Through Tubing Gravel Pack (TTGP)	Through Tubing Sand Screen (TTSS)	Sand Treatment	Passive Oriented Perforation
Production Benefit	Effective sand control	Practical sand retention	Provides more drawdown	Provides more drawdown
	High flow potential	Medium flow potential	Effectiveness depends on chemical success	High flow potential
Economics	Expensive gravel placement	Cost-effective	Expensive chemicals cost + vessel support	Cost-effective
	Conveyance via SL & CTU	Conveyance via SL	Conveyance via CTU & Pumping Unit	Conveyance via E-Line
Estimated Cost (Range)	High	Low to Medium	Very High	Low to Medium
Practicality	Complex retrieval process	Quick wireline installation	Requires CTU for effective treatment & cleaning	Deployment simplicity
	Slow installation	Easy remedial action	Complex operation	Proven deployment
Operational Duration (approx.)	7 – 9 days	3 days	7 – 14 days	5 days
Reliability	Proven sand control method	Screen potential plugging	Highly sensitive to reservoir condition, length and permeability contrast	Proven deployment & application
	Long term effectiveness	High mechanical strength	Must obtain good sand retention test result	Proven sand control method
Track Record	Good performance	Fair performance	Promising so far	Well-establish performance

Geomechanics study and Historical Data

A comprehensive geomechanical feasibility study was undertaken to evaluate the suitability of optimum perforation method for SK Gas well with cemented monobore completion type. The study involved analyzing in-situ stress profiles, pore pressures, and rock mechanics data derived from well logs, core samples, and pressure tests. This allowed for the precise identification of optimal perforation methods. Implementing oriented perforation requires advanced wellbore analysis, including geomechanically modeling and reservoir stress studies, to determine the optimal perforation angles and depths. These techniques adjust wellbore parameters, pressure conditions, and fluid flow dynamics to minimize the risk of sand migration and production, allowing operators to avoid the costs and complexities associated with mechanical or chemical solutions. The importance of geomechanical analysis (data acquisition & analysis) is described in SPE paper 86470 (J. Tronvoll et al, 2004).

The workflow of Geomechanical study done in-house is shown in [Figure 3](#).

An Operating Envelope was constructed based on the failure model for a specific perforation angle and depth. The red line represents the Critical Bottom Hole Pressure (CBHP), while the green line denotes the reservoir pressure. The intersection of these lines, referred to as P_{fail} , marks the point below which sand production is assumed.

Furthermore, the Maximum Drawdown Limit (the gap between CBHP and reservoir pressure) which defines the safe drawdown to produce without sand issues will give a better caution on the drawdown allowance limit. As the reservoir pressure depletes, the drawdown limit decreases, ensuring safe production without excessive sand intrusion.

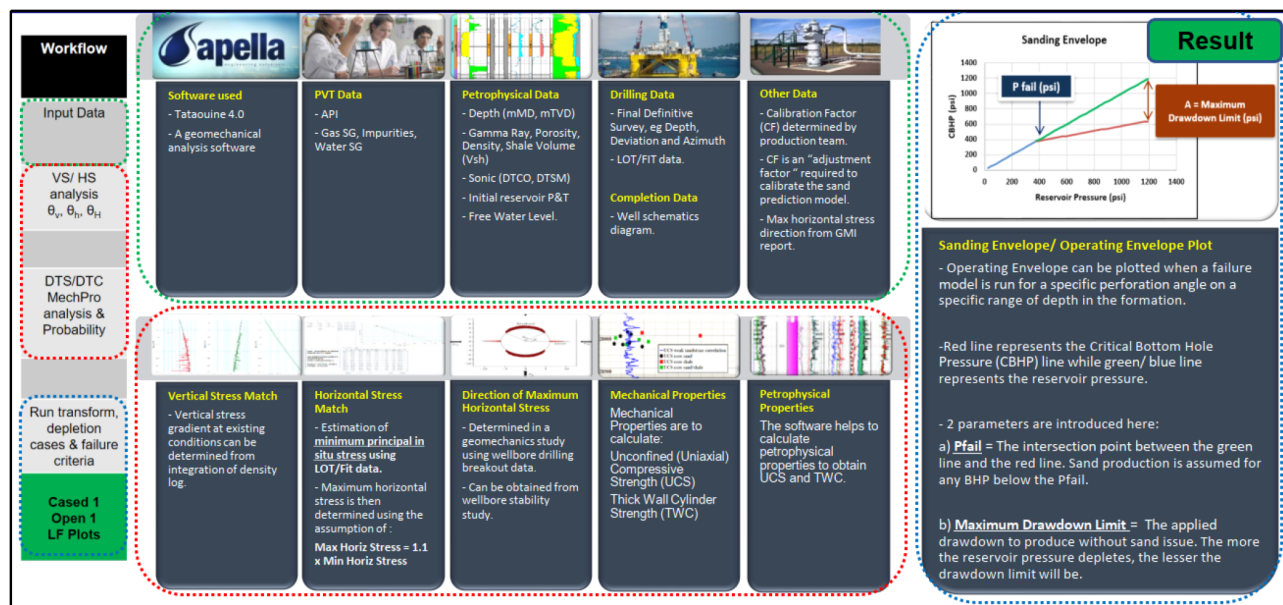


Figure 3—Workflow of Geomechanical Study

The example of geo-mechanical results for 3 well-recompletion candidates shown in Figure 4.

Well Name	Deviation @ Perforated Layer (Deg)	Perforated Sand Layer	Length (mMD)	Top Perf (m TVDss)	Bottom Perf (m TVDss)	Oriented Perforation Pfail Gain (psi)	Oriented Perforation Drawdown Gain (psi)
Well E	56.1	Sand D	4.0	1205.4	1208.6	+800	+350
Well F	23.4	Sand E	8.0	1186.1	1193.5	+220	+130
	24.0	Sand F	6.4	1209.3	1215.3	+215	+140
Well G	72.4	Sand G	3.5	808.0	809.1	+370	+380

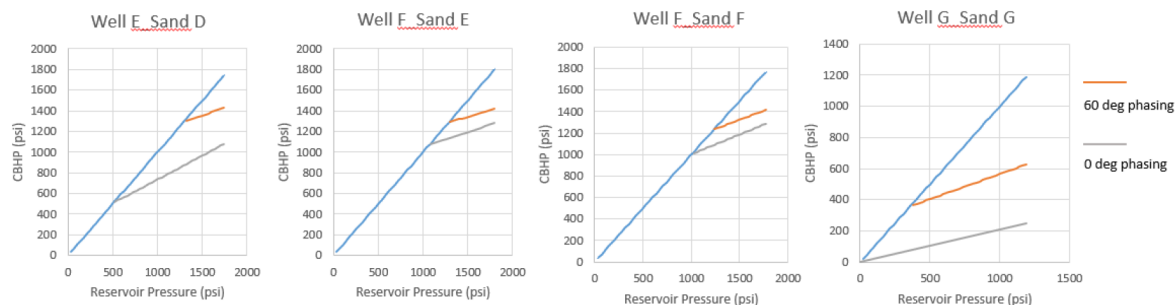


Figure 4—Sanding Envelope / Operating Envelope

Based on Geomechanics study, to change from Standard Perforation to Oriented Perforation while managing the drawdown efficiently is the key to reducing or managing sand production without physically excluding sand from entering the wellbore. Based on result, the 0-deg gun phasing gives more drawdown limit.

Oriented Perforation Journey

An oriented perforation technique is a strategic method used to control sand production by carefully designing the direction and placement of perforation tunnels in the wellbore. The preference is to align these perforations in the direction of maximum horizontal stress of the formation, ensuring perforation tunnel stability and reducing the likelihood of sand production. This method aims to maximize the perforation tunnel strength and minimize flow restrictions. By achieving stable perforation tunnels, the risk of sand

influx and collapse is significantly reduced, even in weakly consolidated or unconsolidated reservoirs. SPE paper 93639 (A.J Martin et al, 2005) emphasizes the accurate oriented perforation job will benefit in extending the well productivity (minimize / delay sand production). The illustration of oriented perforation as in Figure 5.

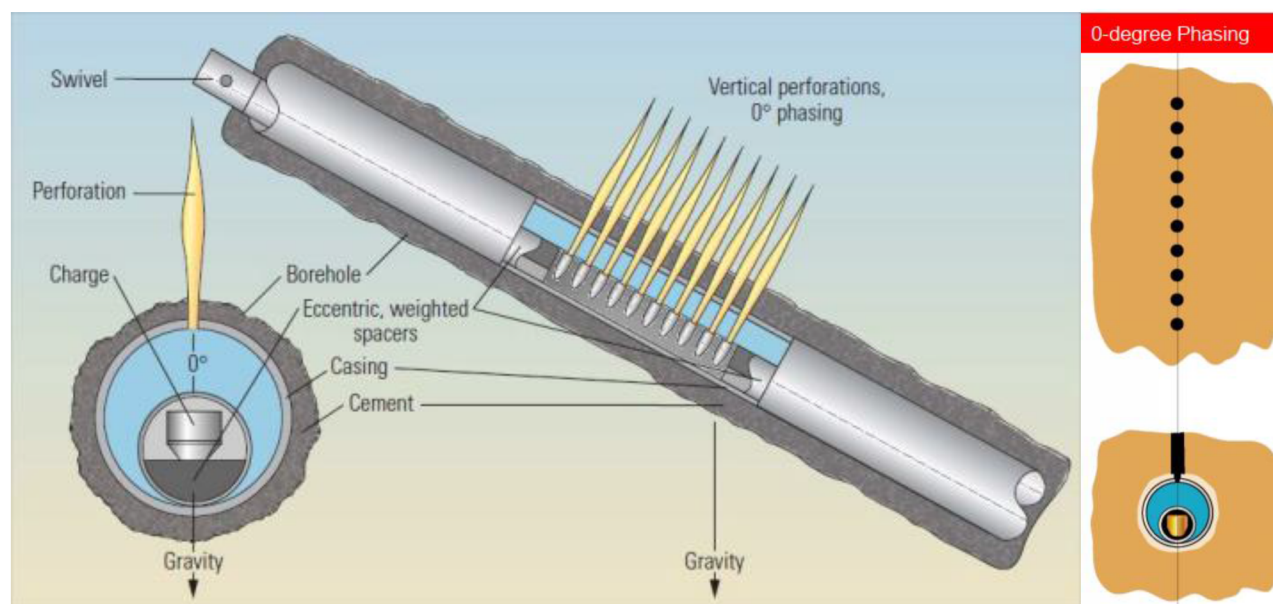


Figure 5—Oriented Perforation design

There are 2 types of Oriented Perforation Technique which is active oriented perforation and passive oriented perforation evaluated.

- An active oriented perforation typically relies on gyroscopic or motorized device to direct charges toward specific reservoir zones. While effective in many scenarios, these systems can be expensive, complex, and sensitive to wellbore conditions. In cemented monobore wells, achieving reliable orientation is particularly difficult due to limited tool maneuverability and alignment constraints.
- A passive oriented perforation represents a shift from active control to natural alignment mechanisms. These systems utilize gravity, tool asymmetry, and wellbore geometry to achieve consistent orientation without external inputs. An orienting weight (one sided weight) added to the selected carrier and with a swivel located at the top to make the BHA rotate freely. The BHA will orientate based on the weight-gravity effect to desired (0 & 180-deg phasing). This approach reduces complexity, enhances reliability, and is well-suited for cemented monobore wells where simplicity and robustness are essential.

For SK Gas well recompletion candidates, the method selected is Passive Oriented Perforation. The technique was selected primarily due to its availability, simplicity and cost efficiency. The oriented perforation technique required a specialized gun carrier with oriented weight bar design to ensure proper alignment and orientate the perforation tunnel parallel to the highest in-situ stress plane. The tool incorporated a one-sided weight to enable gravity-assisted orientation, and a swivel mechanism was added to allow the bottom hole assembly (BHA) to rotate freely. Perforation tunnels were configured with 0° and 180° phasing, with 4 shots per foot, to create selective upward and downward perforations in separate runs and details are captured on Field Execution section.

- First run, the gun charges are aligned in 0 deg direction, with the weight bar. When the weight bar falls under gravity, firing the gun will cause to shoot upwards (0 deg).

- Second run, on surface, the gun carrier is now rotated to 180 deg relative to the weight bar. When the weight bar falls under gravity, firing the gun will cause to shoot downwards (180 deg).

Perforation Simulation

The perforation simulation plays important roles in selecting the most suitable gun system for each well candidate. By integrating formation and wellbore parameter – such as reservoir pressure, UCS, casing size, and expected production rates – the simulation helps engineers determine the optimal perforation strategy to achieve desired gas production rates. Parameters such as shot density, phasing, penetration depth, and charge type are evaluated to ensure effective reservoir connectivity. Another key aspect of the simulation is the evaluation of Dynamic Underbalance, which includes:

- Influx Rate: The rate at which formation fluids enter the wellbore post-perforation, influencing cleanup efficiency.
- Tool Lift Force: The upward force exerted on the perforating gun due to pressure differential, which can cause gun jump or misalignment.

Details guidelines on perforation recommendation on gun system selection are discussed in SPE paper 57954 (A.L Sulbaran et al, 1999) which remains as important reference in perforation planning. Figure 6 shows the direction of the perforation tunnel (0-deg & 180-deg phasing) and Figure 7 shows the perforation simulation result in selecting optimal gun system (charges type, shot density, AOF, flow rate).

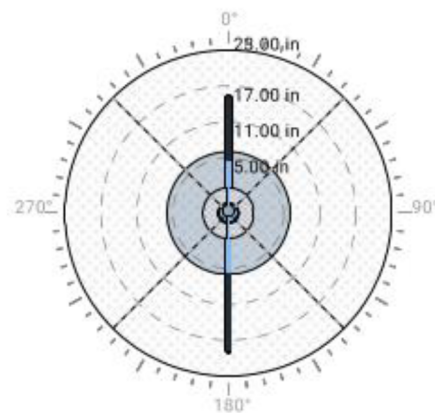


Figure 6—0 & 180-deg phasing (max horizontal stress)

Perf #	Loaded Length (m)	Phasing Angle (deg)	Shot Density / Open Perfs (spf/%)	Eff Shot Density (spf)	Clean Length Lc/L	Form Pen Avg (in)	Form Dia Avg (in)	EH Dia Avg (in)
1	6.4	180	4.00/100	4.00	0.33	14.01 *	0.68	0.20
2	6.4	180	8.00/100	8.00	0.33	14.01 *	0.68	0.20

Perf #	Eff Skin	Darcy Skin	Perf Skin	Crush Zone Skin	Deviation Skin	Partially Open Skin	Non-Darcy Coeff (1/MMSCF/day)	Non-Darcy Skin	PR	PI * (MMSCF/day/psi)	Flow * Rate (MMSCF/day)
1	12.07	7.66	8.27	0.00	-0.61	0.00	0.8828	4.405	0.47	0.05	4.99
2	10.61	6.14	6.75	0.00	-0.61	0.00	0.8261	4.469	0.52	0.05	5.4094

Figure 7—Perforation simulation output

System Integration Testing (SIT) & Field Execution of Passive Oriented Perforation

A System Integration Testing (SIT) was done at service provider Labuan warehouse to evaluate on the following:

- Visual Inspection & Connection Compatibility

- All oriented weight bars were visually inspected.
- Each bar was made up with the gun adapter to verify connection compatibility and ensure proper fitment.
- Charge Azimuth Verification
 - The oriented bar was assembled with a 2" HSD gun.
 - Ground-level rotation tests were performed to verify charge azimuth alignment for 0° and 180° phasing, ensuring correct orientation for perforation shooting.

Table 3—SIT Summary

SIT	Objective	Methodology & Result
#1 in Q4 2020	- To perform physical connection check between 10ft of 2" Oriented weight bar and gun adapters. - To make up 20ft of 2" HSD gun with 10ft of 2" Oriented weight bar and simulate shooting with charges at 0-deg and 180-deg phasing.	- Fit test the oriented weight bar with gun adapter. Checked physical presence of half-moon weight. - Make up gun string in 2 configurations #1: Shot at 0-deg #2: Shot at 180-deg - The gun was rolled on horizontal area. - The oriented weight was able to orient the 20ft gun above it and shot aligned to 0-deg or 180-deg as expected - Video was recorded and shared to show the simulation
#2 in Q3 2022	- To perform physical connection check between 15ft of 2" Oriented weight bar and gun adapters. - To make up 30ft of 2" HSD gun with 15ft of 2" Oriented weight bar and simulate shooting with charges at 0-deg and 180-deg phasing.	- Fit test the oriented weight bar with gun adapter. Checked physical presence of half-moon weight. - Make up gun string in 2 configurations #1: Shot at 0-deg #2: Shot at 180-deg - The gun was rolled on horizontal area. - The oriented weight was able to orient the 30ft gun above it and shot aligned to 0-deg or 180-deg as expected - Video was recorded and shared to show the simulation

On the first 2 years (2021 & 2022), a total 20 wells were successfully implemented with passive oriented perforation as part of SK Gas remedial sand management strategy. The capability of passive oriented perforation is limited to:

- 10ft Oriented Weight Bar (limited to 20ft gun per run)
- Selective perforation at 0 & 180 deg gun phasing
- Separate run for each phasing
- Tractor used for high angle well

With the nature of SK Asset Field operating mode – daily tripping operation, the team starts to explore for an optimization on overall perforation operation efficiency. By having a 15ft oriented weight bar and doability to make single run for both 0 & 180deg gun phasing, the capability now expands to:

- 15ft Oriented Weight Bar (limited to 40ft gun per run)
- Selective perforation at 0 & 180 deg gun phasing

- Single run for 0 & 180 deg gun phasing
- Roller boogies introduced for high angle well
- Oriented perforation integrated with D-BOP system

The transition from separate runs to single-run dual-phasing setups significantly reduced operational time and improved operational efficiency, especially for daily tripping operation mode in SK Asset field. The integration of conveyance technologies and orientation systems further enhanced deployment reliability in complex well geometries. Another 20 more wells were successfully done within 2023 up to Q2 2025.

Refer to [Table 4](#) for the year-by-year execution summary.

Table 4—Oriented Perforation Execution Summary year-by-year

Years	2021	2022	2023	2024	2025
No of Wells	6	14	10	9	1

The following configurations illustrate various field-proven applications of passive oriented perforation using different gun lengths, phasing types, and oriented weight bar lengths. These examples in [Table 5](#) demonstrate the method's flexibility in adapting to different reservoir targets and operational objectives:

Table 5—Oriented Perforation Gun Configuration

Configuration Type	Gun Length	Phasing	Oriented Weight Bar Length
Single Gun	10 ft	Single Phasing	10 ft
Single Gun	20 ft	Single Phasing	10 ft
Selective Gun	10 ft + 10 ft	Single Phasing	10 ft
Selective Gun	10 ft + 10 ft	Dual Phasing	10 ft
Selective Gun	20 ft + 20 ft	Single Phasing	15 ft
Selective Gun	20 ft + 20 ft	Dual Phasing	15 ft
Tandem Gun	10 ft + 10 ft	Single Phasing	10 ft
Tandem Gun	10 ft + 10 ft	Dual Phasing	10 ft
Tandem Gun	20 ft + 20 ft	Single Phasing	15 ft
Tandem Gun	20 ft + 20 ft	Dual Phasing	15 ft

Results and Discussion

Operational Efficiency

The deployment of passive oriented perforation in SK Gas well with cemented monobore completion type proved to be a technically and economically effective solution for remedial sand management and production enhancement in cemented monobore well. Over the course of the campaign:

- No SSHE Incidents for 40 wells execution
- All gas Producer candidates were completed using passive oriented perforation at 0 & 180-deg phasing.
- 259 perforation runs were successfully executed.
 - Only 8 misfires were recorded, indicating a high reliability rate of over 97%.

- The use of gravity-assisted orientation tools, combined with advanced conveyance systems like roller bogies and tractors, enabled conveyance and precise perforation even in high-deviation wells.
 - 8 wells conveyance via wireline Tractor & 14 wells conveyance via Roller Boogies
- 5 wells using Deployment BOP system
 - Minimize the number of runs for long perforation guns in highly deviated wells

Post-Production Performance

Post-job MSFR testing revealed that wells could sustain higher drawdown limits without exceeding sand-free thresholds. This allowed for:

- Increased production rates, with improved reservoir connectivity.
- Reduced need for artificial sand control measures, such as gravel packing or chemical consolidation to avoid unnecessary mechanical skin.
- No significant sand production was observed at surface post-perforation and throughout the production phase to date as summarized in [Table 6](#).
 - This confirming the method's effectiveness in minimizing formation damage and controlling sand influx.
 - As a result, the requirement for well intervention related to sand bailing operations has been significantly reduced.
 - The primary production challenge has shifted towards managing liquid loading and reservoir depletion.
 - An additional intervention was conducted to identify the water-producing zone, further supporting on next production optimization efforts.
 - The surface operation philosophy remains in place, which prohibits the flow of sand to surface. As such, the well is choked back to reduce gas rates and control sand influx accordingly.

Table 6—Post-Production Performance on Oriented Perforation Wells

Production Period	No of Well	Remarks
< 0.5 year	3	● 2 wells unable to sustain in longer period due to reservoir depleted. ● 1 well completed in Q2 2025 and still producing to date.
0.5 - 1 year	5	● 2 wells unable to sustain in longer period due to reservoir depleted ● 1 well completed in Q2 2025 and still producing
1 - 3 years	23	● 14 wells still producing strong to date. ● 5 wells are idle due to liquid loading issue ● 2 wells are currently idle and to conduct isolation on watered out zone to revive production ● 1 well unable to sustain in longer period due to reservoir depleted ● 1 well was SWAP (pressure depleted and no behind casing opportunity) and became a donor well for Infill Drilling
3 years	9	● 3 wells still producing strong above its True Potential (TP) ● 5 wells still producing strong at its True Potential (TP) ● 1 well still producing but below its True Potential (TP) - signed of reservoir depleted

Future Work Recommendations

- **Tool Design Enhancement:** Explore the feasibility of implementing 0° and 180° phasing perforations simultaneously to improve perforation efficiency and sand control performance
- **Integrate Real-Time Logging Capabilities:** Introducing a logging tool within the same perforation run to measure sand production in-situ, and / or confirm the shot placement accuracy within +/- 10° tolerance.
- Iterative improvements in tool design and deployment strategy.
- **Pursue Iterative Design and Deployment Improvements:** Continuously refine the tool design and deployment strategy based on field performance data and lessons learned to enhance operational reliability and effectiveness.

Conclusion

Oriented perforation has demonstrated to be a practical and effective remedial sand control solution for slim-hole cemented monobore wells in the SK Gas Field, successfully addressing the sand management challenges faced by SK Asset since 2020. By aligning perforations with the maximum in-situ stress plane, the technique reduces sand intrusion, enhances drawdown limits, and maximizes production rates. The field results validate its feasibility, highlighting its potential for broader application in similar reservoir conditions.

This journey underscores the importance of innovation and adaptability in addressing sand control challenges. Passive oriented perforation minimizes the need for costly interventions and extends well life, improving overall economic returns. Moving forward, long-term performance monitoring and further advancements in tool design will be crucial to enhancing the reliability and efficiency of this method. The findings of this study contribute valuable insights to the development of sustainable and cost-effective sand management strategies for marginal reservoirs.

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References

- SPE 57954 Oriented Perforation for Sand Prevention. A.L Sulbaran et al Presented at the European Formation Damage Conference, The Hague, 31 May - 1 June 1999
- SPE 86470 The effect of Oriented Perforating as a Sand Control Method: A Field Case Study from the Varg Field. J. Tronvoll et al Presented at SPE International Symposium on Formation Damage Control, Lafayette, 18 - 20 February 2005
- SPE 93639 High Accuracy Oriented Perforating Extends Sand Free Production Life of the Andrew Field. A. J Martin et al Presented in Offshore Europe, 6 - 9 September 2005